

RUDRA SHUKLA

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EDUCATION

University of Waterloo

Candidate for B.A.Sc in Mechatronics Engineering

Waterloo, ON

September 2025 – Present

- **Relevant Coursework:** Data Structure and Algorithms, Engineering Graphics and Design, Circuits, Statics
- **Design Team:** WATonomous (Humanoid Team)

TECHNICAL SKILLS

Languages: Java, Python, C/C++, HTML/CSS, JavaScript

Hardware: ESP32, Arduinos, Circuit Design, Raspberry Pi, Soldering

Tools: OpenCV, AutoCAD, SolidWorks, Soldering, 3D Printing, TensorFlow, Linux, Microsoft Office, PLC, FoxGlove

PROJECTS

AutoniMake - AI Robotics Platform

March 2026

Python, C/C++, OpenCV, TensorFlow, Raspberry Pi, ESP32, Embedded Systems, Web Development Waterloo, ON

- Built a code-free AI robotics platform enabling users to collect image data, **train CNN models**, and deploy real-time predictions through a web interface.
- Developed a real-time vision pipeline, and designed a distributed system where a Raspberry Pi handles inference and communicates with ESP32 devices with a custom **UART protocol** for low-latency hardware control.

Automatic Blinds Controller

April 2026

ESP32, C/C++, Embedded Systems, IoT Web Server, 3D Printing, Circuit Design

Ajax, ON

- Built an ESP32-based automatic blinds controller enabling precise bidirectional stepper motor control.
- Developed a **WiFi-enabled web interface** with real-time control (manual, timed, stop), and implemented dual-core tasking, microstepping, and power regulation for smooth, reliable operation.

LED Sign Board

May 2025

ESP32, C/C++, WLED, Soldering, CNC

Ajax, ON

- Designed and built a custom 28x8 LED Sign Board controlled via **IPv4 address** using WLED firmware.
- Fabricated enclosure using **CNC machining** and assembled electronics through wiring and soldering.

EXPERIENCE

Software Quality Analyst

January 2026 – April 2026

i4i (Infrastructures for Information Inc.)

Toronto, ON

- Collaborated with developers throughout the **software development lifecycle** (SDLC) to log, track, and verify bug fixes through structured re-testing, ensuring issues were resolved and aligned with functional requirements.
- Executed detailed test plans using internal tools and structured test cases, **comparing expected vs. actual results** across varied scenarios to identify defects and gaps in functionality.
- Analyzed system outputs and edge-case conditions to uncover inconsistencies and unintended behavior, contributing to more stable and predictable performance in production environments.

Robotics Instructor

January 2023 – August 2025

OBOTZ - IQ Brainers Academy

Ajax, ON

- Developed and delivered a robotics curriculum for **20+** students of varying skill levels, promoting hands-on learning through Arduino-based robot design, programming, and iterative problem-solving exercises.
- Led workshops and competitions, increasing participation by 50% and improving project outcomes.
- Collaborated with industry professionals to **integrate real-world robotics** concepts into STEM programs.

Mechanical and Design Lead - FIRST Tech Challenge

September 2021 – June 2025

J. Clarke Richardson Collegiate

Ajax, ON

- Designed and built robot components using **SolidWorks**, applying structural and performance optimization.
- Led development of **drivetrain, arm, and intake systems** using torque and gear ratio analysis.
- Collaborated with the software team to integrate autonomous functions, **improving scoring consistency by 75%**.

Warrant Officer First Class (WO1) & Senior Instructor

September 2023 – June 2025

856 Royal Canadian Air Cadets

Pickering, ON

- Led training and operations for **200+ cadets**, managing schedules, logistics, and team coordination.
- Conducted leadership and **aviation training, teaching flight theory**, navigation, and aircraft systems.